COMPACT TRAVEL KETTLE DESIGN MILESTONES

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Introduction

Explore how a compact electric kettle can improve the experience of preparing hot drinks in temporary accommodations such as hotels and temporary accommodations.

Focus Areas

Portability
Compact design that can be easily packed during travel.

Hygiene
Personal kettle concept that reduces discomfort when using shared appliances.

User Control
Simple and reliable interaction for preparing hot drinks.

Define: Identifying the Problem

Objective
Explore design directions for a compact travel kettle focused on portability and user confidence.

Approach
Multiple sketches were created to test different shapes, structures, and handle configurations.

Portability
Compact form suitable for packing in travel bags.

Compact Size
Space-efficient design that reduces unnecessary volume.

User Comfort
Simple handle and intuitive interaction for everyday use.

Outcome
Several concept directions that guided the next stage of prototype development

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Prototype: Developing Solutions

Objective
Evaluate the compact travel kettle design and identify improvements.

Key Focus Areas

Usability
Early testing revealed overly complex interaction, leading to a simplified control approach.

Stability
Early testing suggested improvements were needed in the base stability

Portability
Testing highlighted size and weight concerns, resulting in a lighter and more compact form.

Outcome
Prototype testing helped identify key design problems and guided improvements toward a more usable and travel-friendly kettle.

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AI Role in the Design Process

Clarifier
AI helped clarify the problem framing around hygiene concerns and user control when using kettles in temporary accommodations.

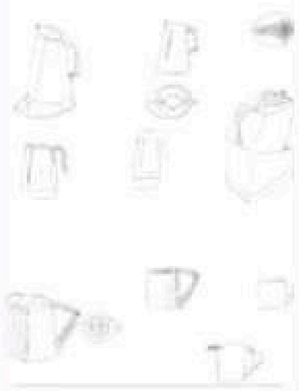
Re-framer
AI suggested alternative ways to view the problem, shifting focus from just portability to user trust and simplicity.

Expander
AI-generated prompts helped explore additional design directions and concept variations during ideation.

Critique
AI responses helped question early ideas and supported reflection on usability, portability, and overall design logic

Design Milestones Draft

01 Research



What problem did you identify?

- Travel kettles are bulky for temporary accommodation
- Shared kettles create hygiene concerns
- Users want simple, compact personal kettles

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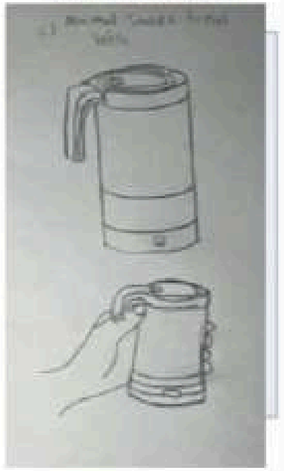
02 Concepts



What directions did you explore?

- Modular detachable kettle
- Collapsible foldable kettle
- Minimal compact kettle

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03 Selection

Why did you choose this concept?

- Minimal cylindrical structure
- Compact and portable form
- Clear and simple user interaction

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04 Prototype

What did the first prototype test?

Overall kettle proportions
- Handle placement
- Structural balance

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05 Problems

What problems Appeared?

- Handle hinge connection unclear
- Rotation axis not visible
- Structural connection confusing

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06 Refinement

What changed after refinement?

Visible hinge axis introduced
- Handle connection clarified
- Structural logic improved